

To test your ALU in the Nand2Tetris Hardware Simulator, you need to set the x and y data inputs, configure the 6 control bits (zx, nx, zy, ny, f, no), and evaluate to see if out, zr, and ng match the expected results.

Here are four essential test cases that verify different parts of your logic, using simple decimal numbers for x and y (the simulator will handle them as 16-bit binary).

Test Cases

Test Goal	x value	y value	Control Bits (zx nx zy ny f no)	Expected out	zr	ng
1. Addition (x + y)	5	3	0 0 0 0 1 0	8	0	0
2. Bitwise AND (x & y)	5	3	0 0 0 0 0 0	1	0	0
3. Force Zero (0)	5	3	1 0 1 0 0 0	0	1	0
4. Subtraction (x - y)	3	5	0 0 0 1 1 1	-2	0	1

Step-by-Step Execution of Test Case 4 (x - y)

Test Case 4 is great for checking your negative flag (ng) because 3 minus 5 results in -2. Here is exactly how your logic routes these bits to arrive at that answer:

1. **zx=0, nx=0:** x passes through unchanged. ($x_2 = 3$)

2. **zy=0, ny=1:** y passes through, but is then bitwise negated. ($y2 = \text{bitwise NOT } 5$)
3. **f=1:** The Add16 gate adds $x2$ and $y2$. ($3 + \text{NOT } 5$)
4. **no=1:** The Mux16 output is bitwise negated. $\text{NOT}(3 + \text{NOT } 5)$ equals mathematically to $3 - 5$.
5. **Output is -2:** In 16-bit two's complement, -2 is 1111111111111110.
6. **Flags:**
 - The leftmost bit (MSB) is 1, so your wire maps $ng = 1$.
 - The output contains 1s, so your Or8Way gates catch them, meaning $zr = 0$.

How to run these in the simulator:

1. Load your ALU.hdl file into the Hardware Simulator.
2. In the **Input pins** section, type 5 for x and 3 for y.
3. Change the control bits (zx, nx, zy, ny, f, no) to match one of the rows above by changing the 0s to 1s where necessary.
4. Click the **Eval** (calculator icon) button.
5. Check the **Output pins** section to ensure out, zr, and ng perfectly match the table.